

Section 1: Introduction to Energy Sources

1. **Energy sources that can be replenished naturally over a short period are called:**
 - A) Non-renewable
 - **B) Renewable**
 - C) Fossil fuels
 - D) Nuclear energy
2. **Which of the following is a non-renewable energy source?**
 - A) Solar
 - B) Wind
 - **C) Natural Gas**
 - D) Biomass
3. **The primary source of almost all energy on Earth is:**
 - **A) The Sun**
 - B) The Moon
 - C) Coal
 - D) Volcanoes
4. **Renewable energy is also known as:**
 - A) Dirty energy
 - **B) Clean or Green energy**
 - C) Expensive energy
 - D) Ancient energy
5. **Which of these is the main cause of global warming?**
 - A) Solar panels
 - **B) Burning fossil fuels**
 - C) Windmills
 - D) Hydroelectric dams

Section 2: Solar Energy

6. **The device that converts sunlight directly into electricity is a:**
 - **A) Photovoltaic (PV) cell**
 - B) Solar cooker
 - C) Generator
 - D) Turbine
7. **Solar cells are usually made of which semiconductor material?**

- A) Iron
 - B) Copper
 - **C) Silicon**
 - D) Aluminum
8. **A collection of several solar cells connected together is called a:**
- A) Battery
 - **B) Solar Panel (Module)**
 - C) Solar furnace
 - D) Circuit
9. **Solar thermal energy is used primarily for:**
- A) Lighting bulbs
 - **B) Heating water or air**
 - C) Running computers
 - D) Charging phones
10. **The main disadvantage of solar energy is that it:**
- A) Produces noise
 - B) Causes pollution
 - **C) Is not available at night or during cloudy weather**
 - D) Is very heavy

Section 3: Wind and Hydroelectric Energy

11. **Wind energy is the kinetic energy of:**
- A) Water
 - **B) Moving air**
 - C) Sunlight
 - D) Steam
12. **The structure used to harness wind energy is a:**
- A) Solar tower
 - **B) Wind Turbine**
 - C) Dam
 - D) Piston
13. **Hydroelectric energy is produced from:**
- A) Static water
 - **B) Falling or flowing water**
 - C) Boiling water
 - D) Rainwater only

14. In a hydroelectric power plant, the kinetic energy of water rotates a:
- A) Solar panel
 - **B) Turbine**
 - C) Mirror
 - D) Boiler
15. The largest hydroelectric dam in Pakistan is:
- **A) Tarbela Dam**
 - B) Mangla Dam
 - C) Warsak Dam
 - D) Rawal Dam

Section 4: Biomass and Geothermal Energy

16. Organic matter like wood, agricultural waste, and animal dung used as fuel is:
- A) Fossil fuel
 - **B) Biomass**
 - C) Geothermal
 - D) Nuclear
17. Biogas is primarily composed of which gas?
- A) Oxygen
 - **B) Methane (CH₄)**
 - C) Carbon dioxide
 - D) Nitrogen
18. Energy derived from the heat stored inside the Earth is:
- A) Solar
 - B) Wind
 - **C) Geothermal**
 - D) Tidal
19. Hot springs and geysers are natural sources of:
- A) Hydro energy
 - **B) Geothermal energy**
 - C) Biomass
 - D) Nuclear energy
20. Ethanol, a renewable liquid fuel, is often produced from:
- **A) Sugar cane or Corn**
 - B) Crude oil

- C) Coal mines
- D) Salt water

Section 5: Rapid Fire Review (21-100)

21. **Is Wind energy a form of Solar energy?** (Ans: **Yes, wind is caused by uneven heating of the atmosphere**)
22. **What is 'Tidal Energy'?** (Ans: **Energy from ocean tides**)
23. **Main advantage of renewable energy?** (Ans: **Does not emit CO₂/Infinite supply**)
24. **Device that converts mechanical energy to electrical?** (Ans: **Generator**)
25. **Energy stored in a dam is?** (Ans: **Potential Energy**)
26. **Which country uses the most wind energy?** (Ans: **China/USA/Germany - varies by year**)
27. **What is 'Wave Energy'?** (Ans: **Energy from ocean surface waves**)
28. **Is Nuclear energy renewable?** (Ans: **No, it is finite but low-carbon**)
29. **Can biomass be converted to electricity?** (Ans: **Yes, through combustion or biogas**)
30. **Unit of electrical energy?** (Ans: **Kilowatt-hour - kWh**)
31. **What is a 'Solar Farm'?** (Ans: **Large area with many solar panels**)
32. **What is 'Passive Solar' design?** (Ans: **Using windows/materials to heat buildings naturally**)
33. **Can wind turbines be placed in the ocean?** (Ans: **Yes, Offshore wind farms**)
34. **What is the efficiency of a typical solar cell?** (Ans: **15-20%**)
35. **Does a hydro dam affect fish migration?** (Ans: **Yes, it can block it**)
36. **Greenhouse gas emitted by rotting biomass?** (Ans: **Methane**)
37. **Best region for solar in Pakistan?** (Ans: **Balochistan/Southern Punjab**)
38. **What is 'Gasification'?** (Ans: **Converting biomass into gas fuel**)
39. **Is coal renewable?** (Ans: **No**)

40. **How long does it take for fossil fuels to form? (Ans: Millions of years)**
41. **Main component of a wind turbine blade? (Ans: Fiberglass/Composite)**
42. **Does geothermal energy depend on weather? (Ans: No)**
43. **What are 'Biofuels'? (Ans: Liquid fuels from plants)**
44. **Energy from the rise and fall of sea levels? (Ans: Tidal)**
45. **What is 'Net Metering'? (Ans: Selling excess solar energy back to the grid)**
46. **Environmental impact of Wind energy? (Ans: Noise and bird strikes)**
47. **Can geothermal energy heat homes directly? (Ans: Yes, district heating)**
48. **Source of energy in a 'Solar Cooker'? (Ans: Infrared radiation)**
49. **Most common renewable source in Pakistan? (Ans: Hydroelectric)**
50. **What is 'Sustainability'? (Ans: Meeting current needs without compromising the future)**
51. **Is Wood a renewable resource? (Ans: Yes, if trees are replanted)**
52. **What is 'Photolysis'? (Ans: Splitting water using sunlight to get Hydrogen)**
53. **Are solar panels silent? (Ans: Yes)**
54. **What is a 'Wind Farm'? (Ans: Group of wind turbines)**
55. **Does Hydro energy produce smoke? (Ans: No)**
56. **Source of energy in wood? (Ans: Chemical energy)**
57. **Can energy be created? (Ans: No, only transformed - Law of Conservation)**
58. **Does the moon affect Tidal energy? (Ans: Yes, primarily)**
59. **Is Hydrogen a renewable source? (Ans: Yes, if produced from renewable power)**
60. **What is 'Ocean Thermal Energy Conversion' (OTEC)? (Ans: Using temp differences in ocean water)**
61. **Which renewable source is most reliable? (Ans: Hydro/Geothermal - they are constant)**
62. **Major cost of solar energy? (Ans: Initial installation)**

63. **Does wind speed affect power output?** (Ans: **Yes, significantly**)
64. **Can we use solar energy for cars?** (Ans: **Yes, Solar EVs**)
65. **What is 'Photosynthesis'?** (Ans: **Process plants use to store solar energy**)
66. **Are fossil fuels stores of solar energy?** (Ans: **Yes, ancient solar energy**)
67. **Does geothermal energy produce CO₂?** (Ans: **Very little to none**)
68. **What is 'Pumped Storage Hydro'?** (Ans: **Pumping water up to store energy for later**)
69. **Can garbage be used as energy?** (Ans: **Yes, Waste-to-Energy plants**)
70. **Does solar energy work in winter?** (Ans: **Yes, based on light, not heat**)
71. **What is 'Concentrated Solar Power' (CSP)?** (Ans: **Using mirrors to focus sunlight**)
72. **What is the 'Greenhouse Effect'?** (Ans: **Trapping of heat in atmosphere**)
73. **Does renewable energy improve air quality?** (Ans: **Yes**)
74. **Is Charcoal renewable?** (Ans: **Yes, if from sustainable wood**)
75. **Which is cheaper now: Solar or Coal?** (Ans: **Solar in many regions**)
76. **Can wind turbines interfere with Radar?** (Ans: **Sometimes, yes**)
77. **Is the Earth's core cooling?** (Ans: **Yes, but very slowly**)
78. **What is 'Micro-hydro'?** (Ans: **Small scale hydro for local use**)
79. **What is 'Biodiesel'?** (Ans: **Diesel substitute from veg oils**)
80. **Does renewable energy create jobs?** (Ans: **Yes**)
81. **Can solar panels be recycled?** (Ans: **Yes**)
82. **What is 'Grid Parity'?** (Ans: **When renewable energy costs same as grid power**)
83. **Is sunlight a form of radiation?** (Ans: **Yes**)
84. **What is 'Distributed Generation'?** (Ans: **Generating power near where it's used**)

85. Does global warming reduce hydro power? (Ans: Yes, by causing droughts)
86. Is 'Peat' renewable? (Ans: No, it takes centuries)
87. Can we cook with Biogas? (Ans: Yes)
88. What is 'Anemometer'? (Ans: Instrument to measure wind speed)
89. Does cloud cover stop solar production? (Ans: Reduces it, but doesn't stop it)
90. Can hydro energy be used for irrigation? (Ans: Yes)
91. Is the sun's energy nuclear? (Ans: Yes, Nuclear Fusion)
92. Does wind energy need water? (Ans: No)
93. What is 'Energy Efficiency'? (Ans: Using less energy for same task)
94. Is 'Carbon Footprint' lower with Wind? (Ans: Yes)
95. What is a 'Solar Inverter'? (Ans: Converts DC from panels to AC for home)
96. Does geothermal energy cause earthquakes? (Ans: Rarely, if water is injected at high pressure)
97. Can biomass reduce waste in cities? (Ans: Yes)
98. Is renewable energy the future of Pakistan? (Ans: Yes, to reduce oil imports)
99. What is 'Intermittency'? (Ans: Energy sources that aren't always 'on')
100. The ultimate goal of using renewable energy is? (Ans: Environmental sustainability)